

Image restoration



Acquired image is a degraded version of the original scene

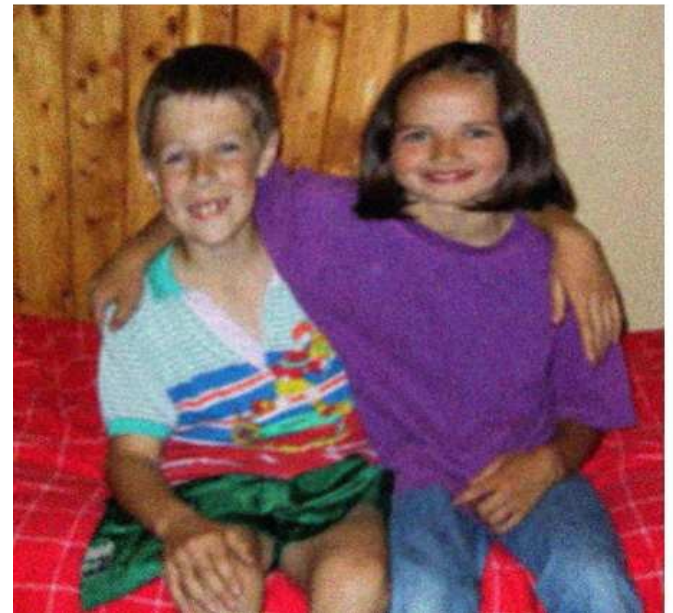


Image degradation model

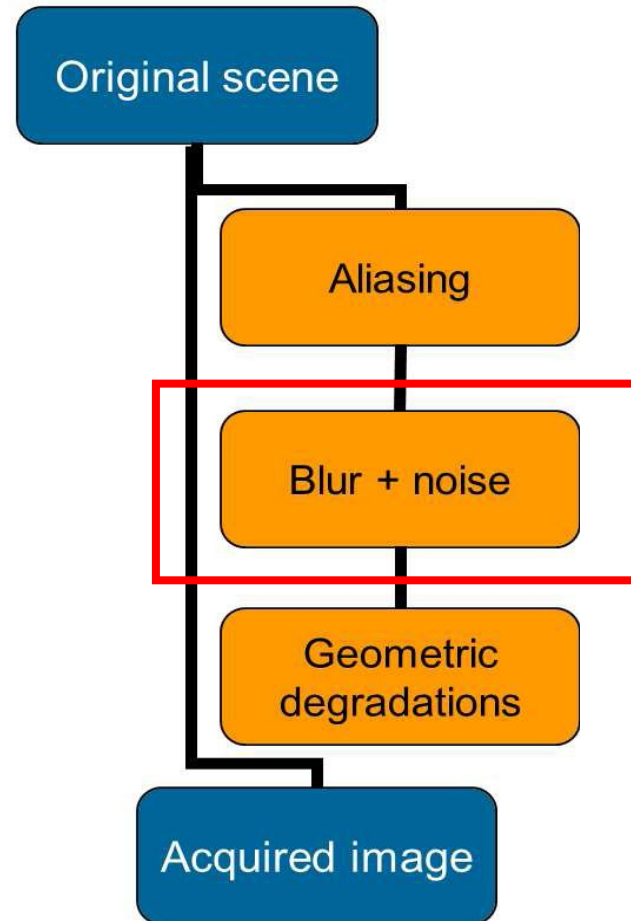
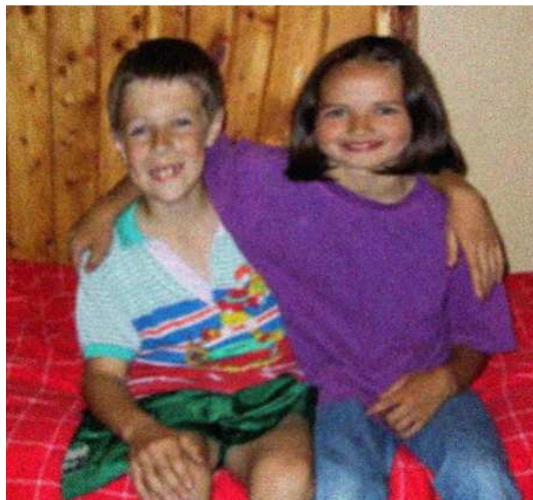
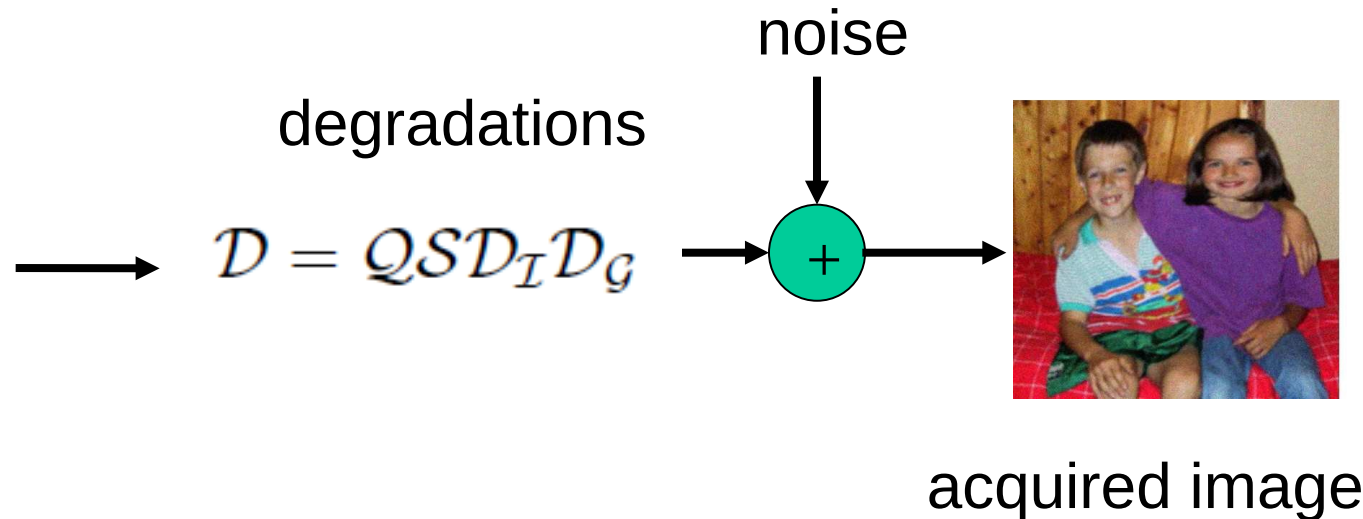


Image acquisition model



original scene



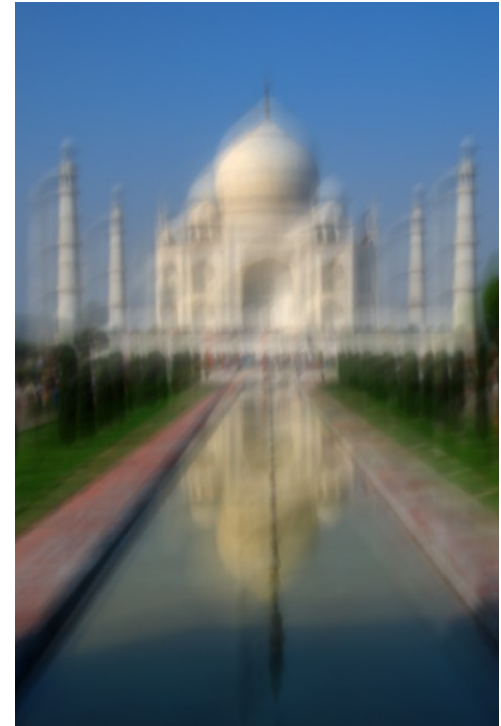
$$\mathcal{D}_G(f)(x, y) = f(\tau(x, y))$$

$$\mathcal{D}_I(f)(x, y) = \int_{\Lambda} \int_T \int \int h(x, y, a, b, \lambda, t) f(a, b) da db d\lambda dt$$

Sources, models and appearance of individual degradations

Blur

Image smoothing, low-pass filter
Suppression of high frequencies



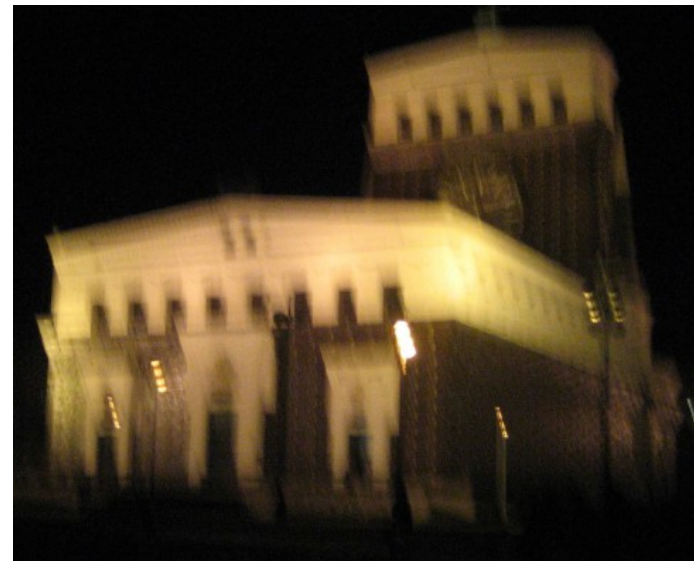
$$\mathcal{D}_I(f)(x, y) = \int_{\Lambda} \int_T \int \int h(x, y, a, b, \lambda, t) f(a, b) da db d\lambda dt$$

Blur examples



Typical blur sources

Camera shake/motion



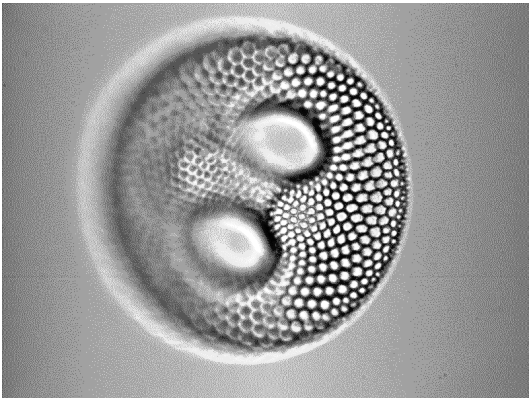
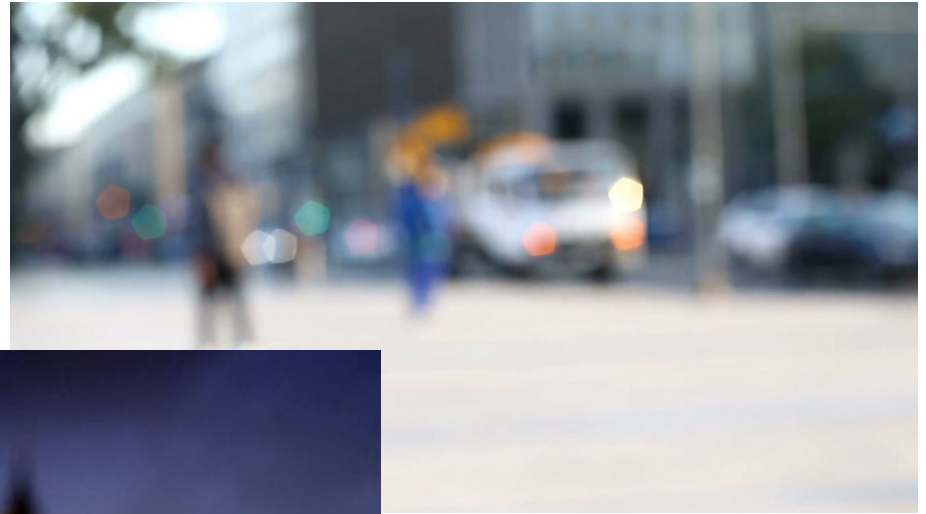
Typical blur sources

Scene/object motion



Typical blur sources

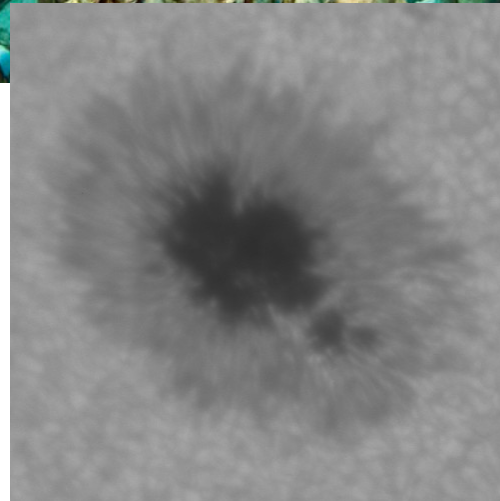
Wrong focus



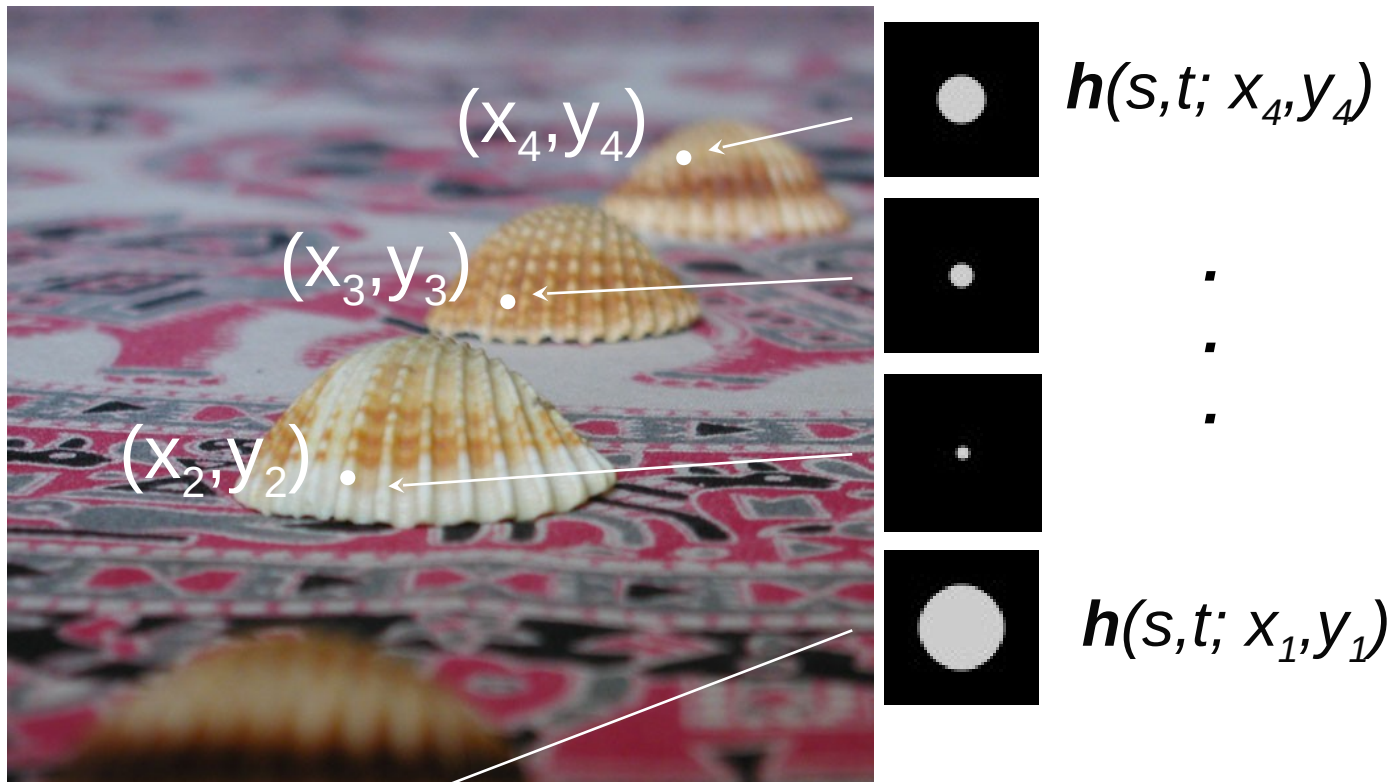
Karen Walrond

Typical blur sources

Medium turbulence



General space-variant blur model



$$\mathbf{z}(x, y) = \int_{\Omega} \mathbf{u}(x - s, y - t) \mathbf{h}(s, t; x - s, y - t) ds dt + \mathbf{n}(x, y)$$

Simplified space-invariant blur model



- *Flat scene*

- *Constant*
- *motion*



$$z(x) = (h * u)(x) + n(x)$$

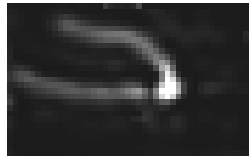
$h(x)$ is the PSF of the camera

Understanding PSF

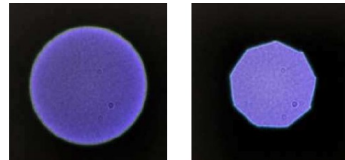
- PSF is the response to
- an ideal point source

$$z(x) = (h * u)(x) + n(x)$$

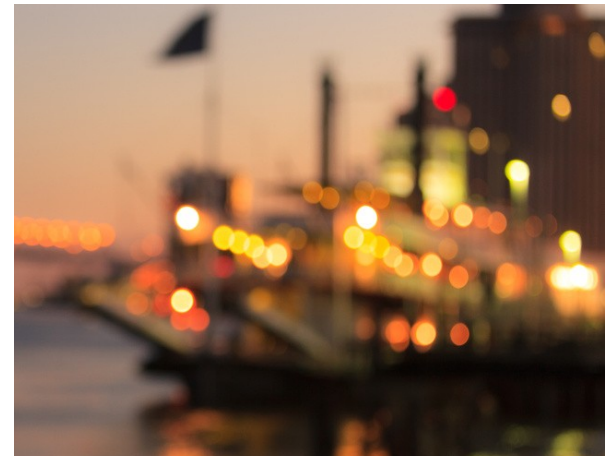
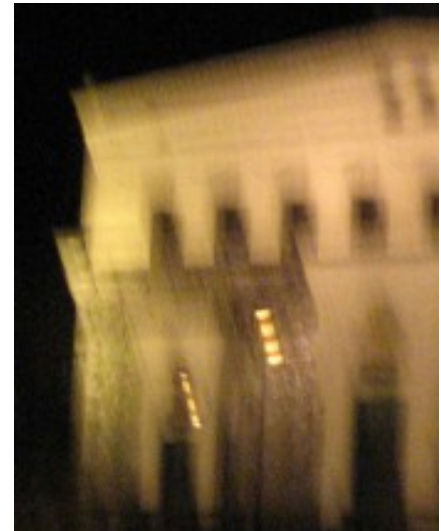
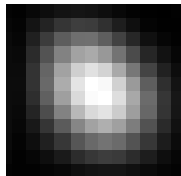
- *camera shake/motion*



- *out-of-focus*



- *turbulence*



Recalling convolution

$$(f * g)(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(s, t) g(x - s, y - t) ds dt$$

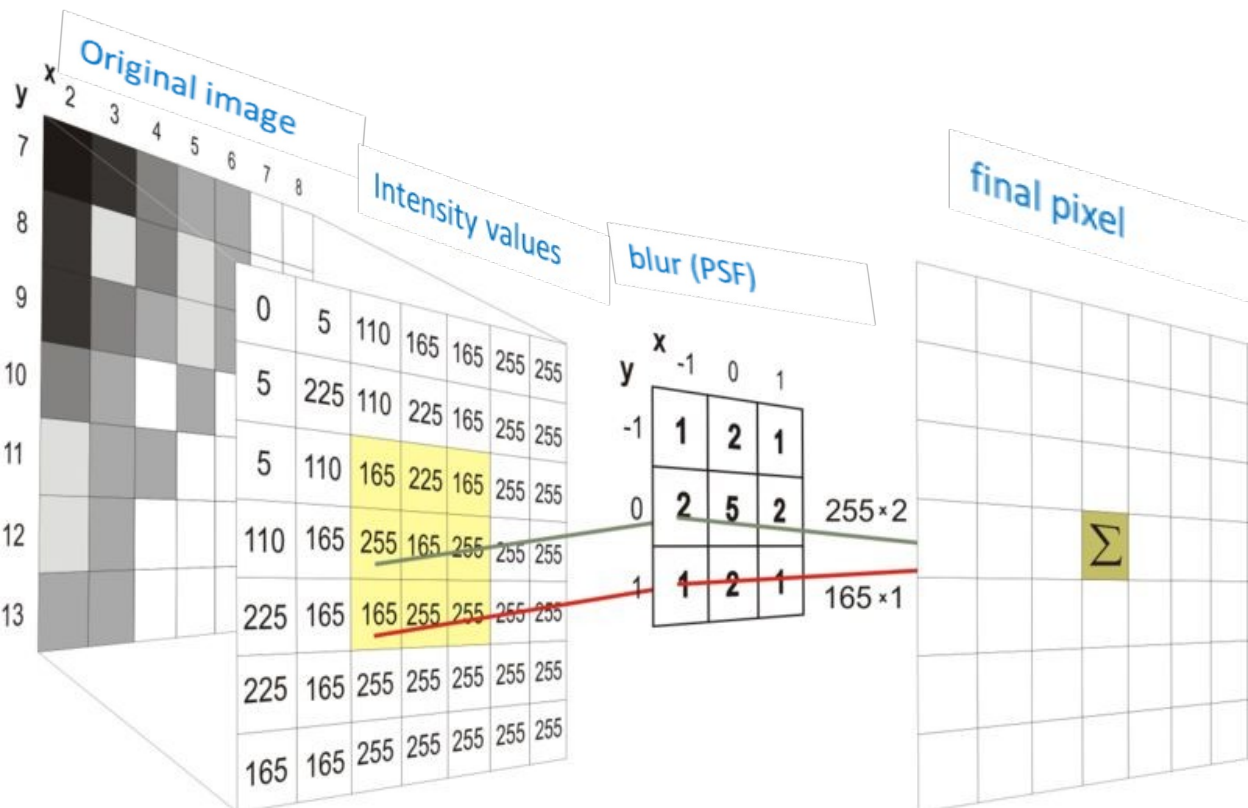
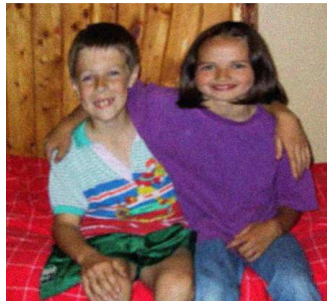


Image restoration

“Inverting” the degradations $g = \mathcal{D}(f) + n$



Looking for $\hat{f}$ such that $\varrho(f, \hat{f})$ is minimized

Noise-free case

$$\hat{f} = \mathcal{D}^{-1}(g) = \mathcal{D}_{\mathcal{G}}^{-1} \mathcal{D}_{\mathcal{I}}^{-1} \mathcal{S}^{-1} \mathcal{Q}^{-1}(g)$$

Noisy case

$$g = \mathcal{D}(f) + n$$

Noise makes the problem ill-conditioned and difficult to handle

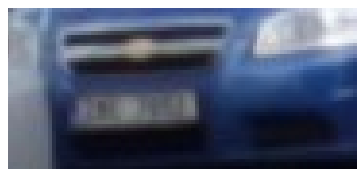
$$\hat{f} \neq \mathcal{D}^{-1}(g)$$

Denoising and/or regularization techniques are required.

Historical remark

- Image restoration has been a very traditional area of image processing
- A. Rosenfeld: *Picture Processing by Computer*, Academic Press, 1969
- 2018 - 200 000 000 search results by Google
- - 3 000 000 results at Google Scholar

Why is image restoration difficult?



- The problem is ill-conditioned and/or ill-posed

$$z(x) = (h * u)(x) + n(x)$$

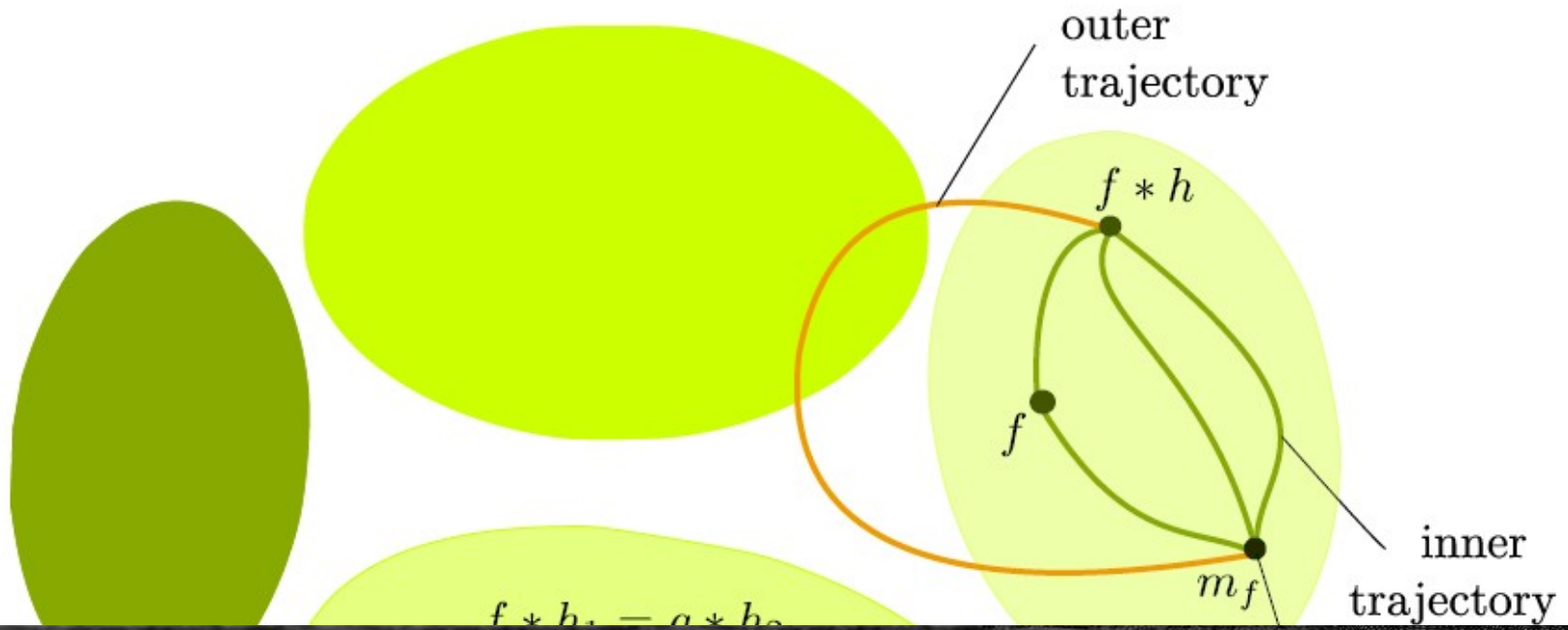
- Too many unknowns
- Even if the PSF is known, the noise makes the task ill-conditioned and requires a special care (regularization)

Why is image restoration difficult?

- Even if we had a perfect deconvolution algorithm and no noise was present, there would be still a solution ambiguity

$$z(x) = ((h_1 * h_2 * \cdots * h_L) * u)(x)$$

Understanding deconvolution



What shall we do?

- We need to choose the correct trajectory in the image space and the point where to stop
- This can be supported by incorporating our preferences/priors/constraints into the restoration algorithms
- Both the original image f and the PSF can be constrained

Restoration categories

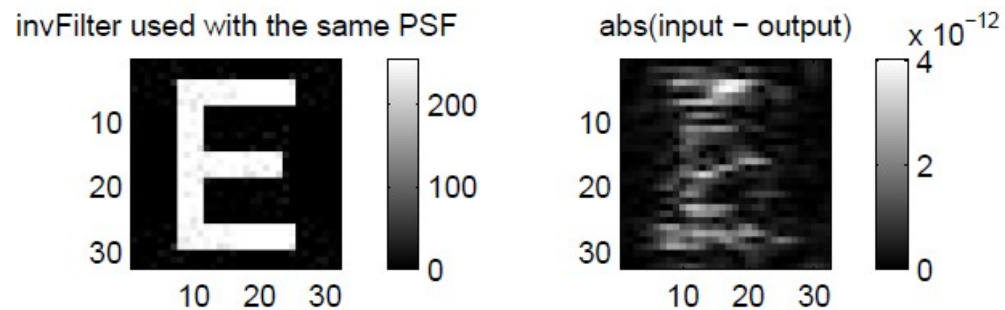
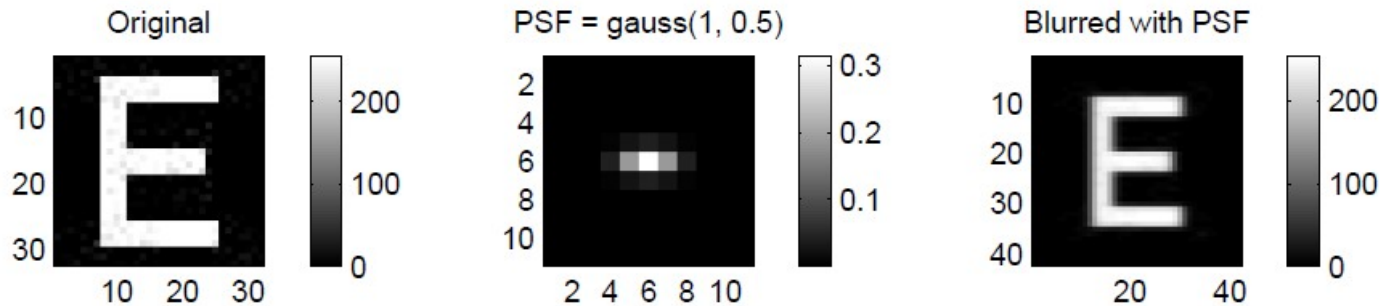
- **PSF is completely known**
- **PSF is constant and of a known parametric shape**
- **PSF is constant and unknown**
- **PSF is variable and unknown**

Intuitive solution to the inverse problem

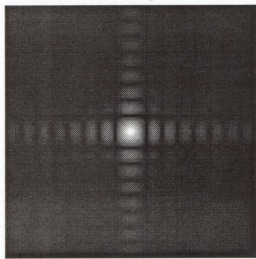
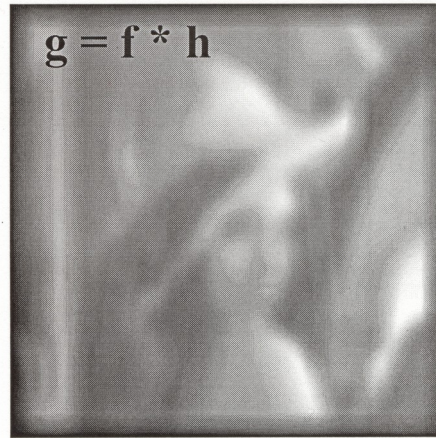
- No noise, PSF known – Fourier transform

$$G = F \cdot H$$

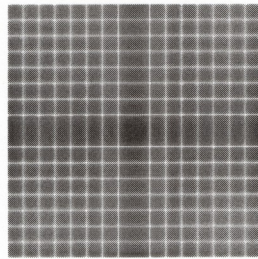
Inverse filter in Fourier domain



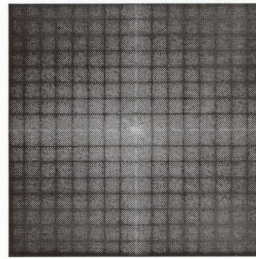
Inverse filter in Fourier domain



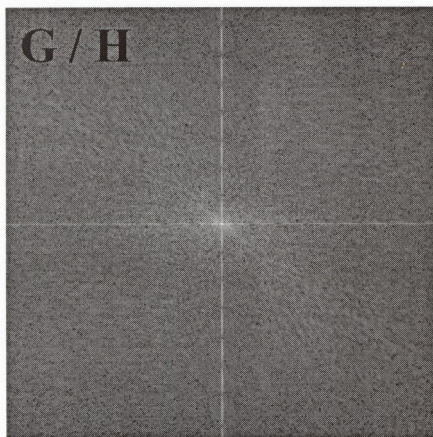
H



$1/H$



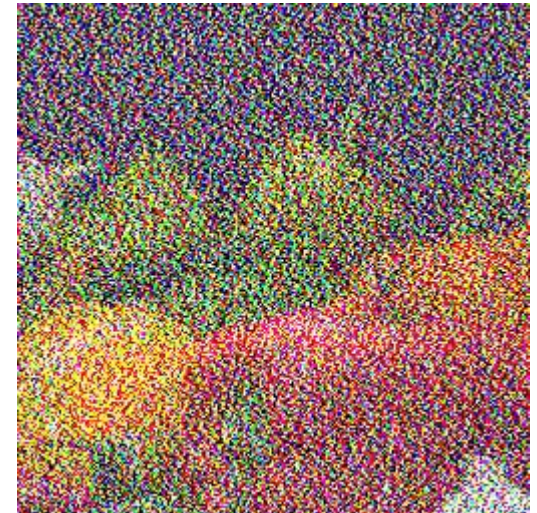
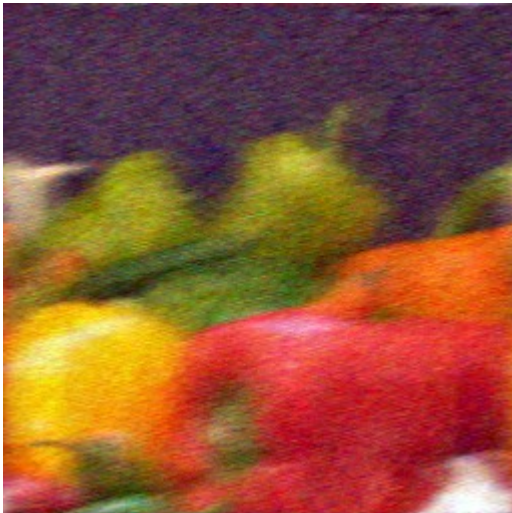
G



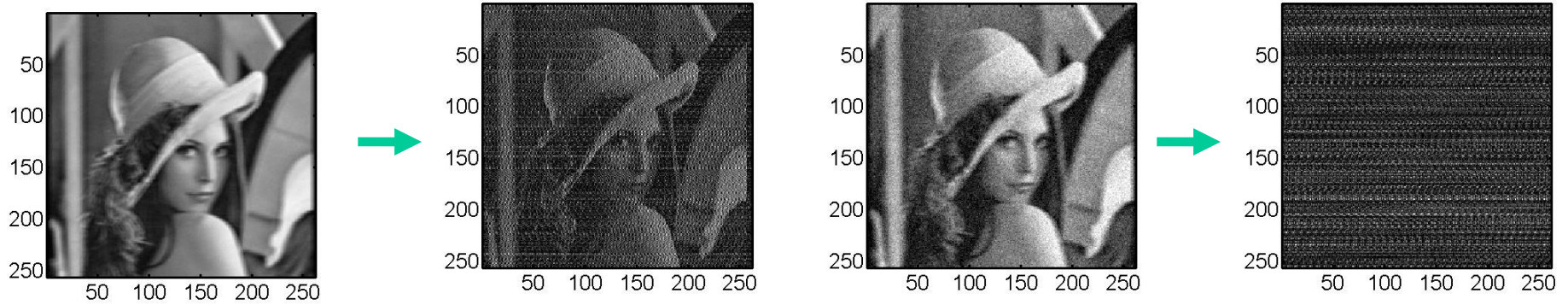
Intuitive solution to the inverse problem

- ... does not work on images with noise

$$G = F \cdot H + N$$
$$F = \frac{G}{H} - \frac{N}{H}$$



- ... does not work on images with noise



Wiener filter

$$E(\|f' - f\|^2) \rightarrow \min$$

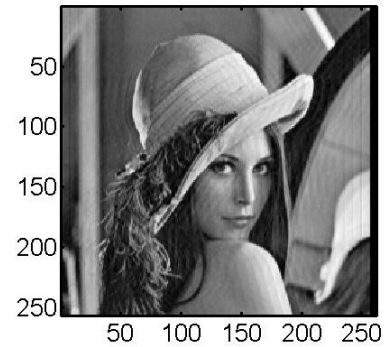
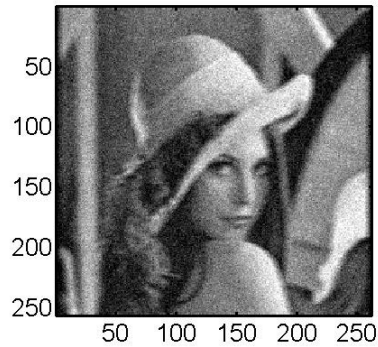
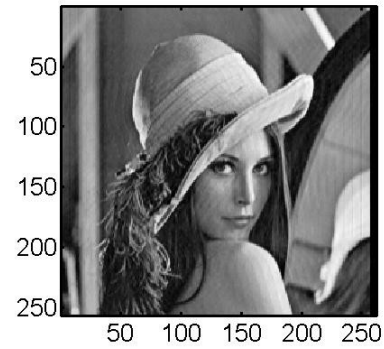
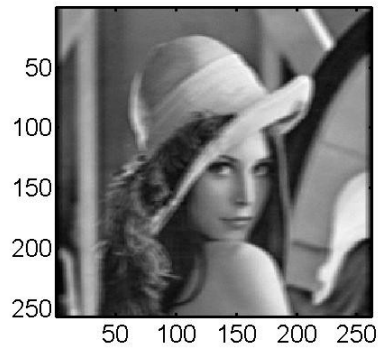
- Minimization on the set of linear filters $F' = RG$

Wiener filter

$$R(u, v) = \frac{1}{H(u, v)} \cdot \frac{|H(u, v)|^2}{|H(u, v)|^2 + S_n(u, v)/S_f(u, v)}$$

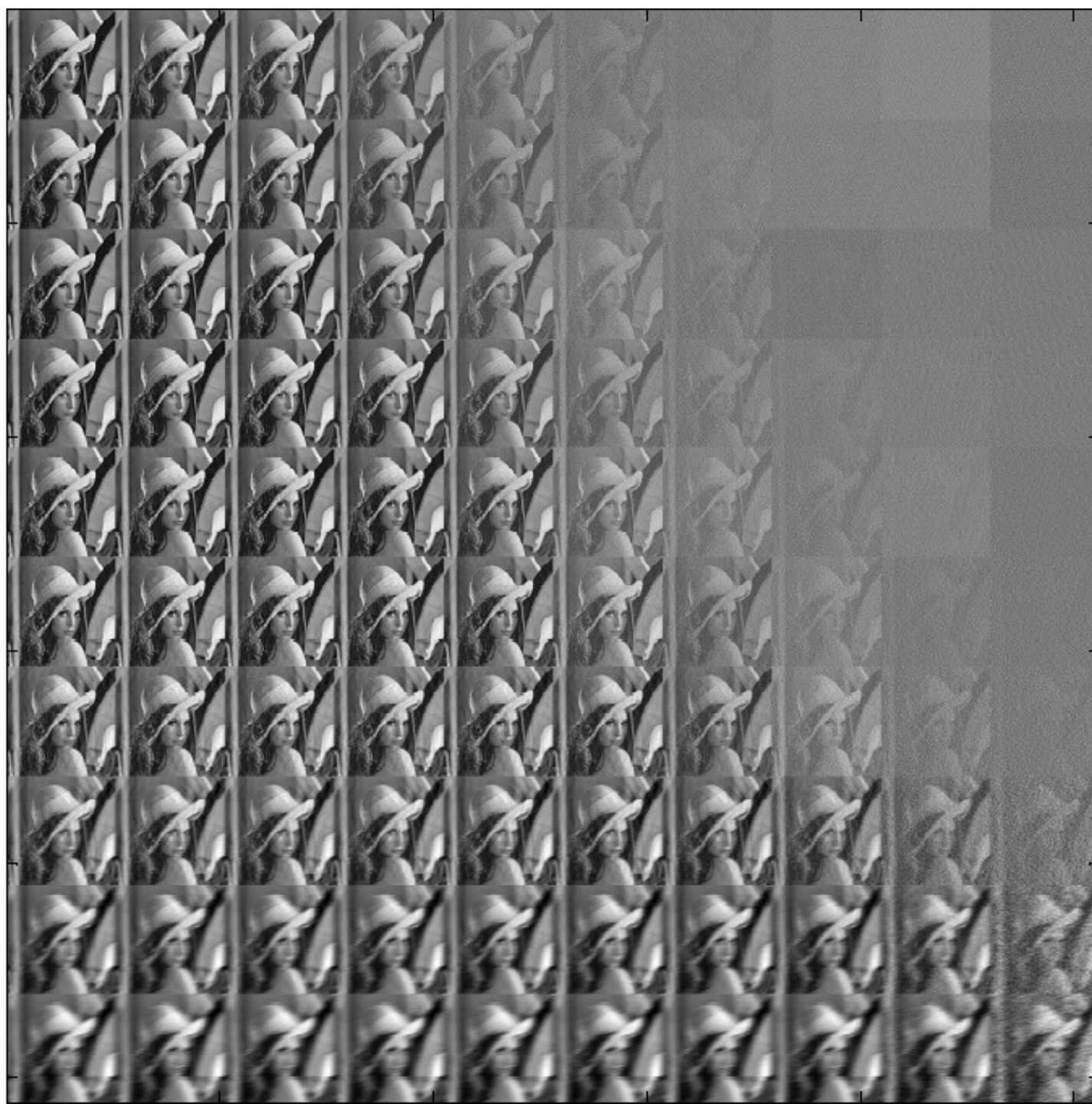
- $S_n(u, v)/S_f(u, v) \approx SNR^{-1}$

Wiener filter



The role of the SNR value





The role of the PSF knowledge



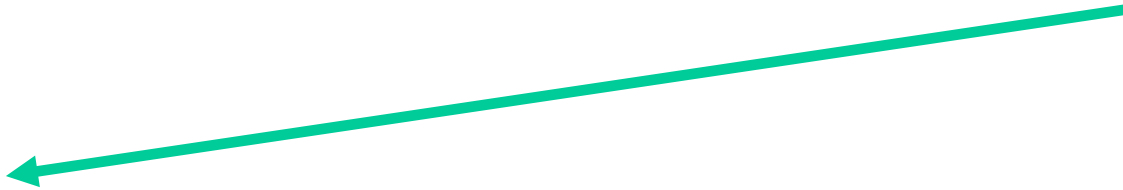
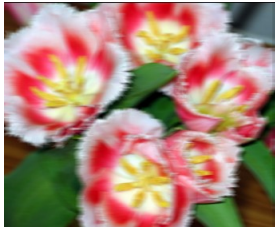


Image restoration flowchart

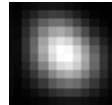


Blurred image

Image restoration flowchart

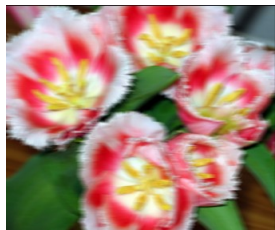


Blurred image

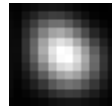


PSF estimation

Image restoration flowchart



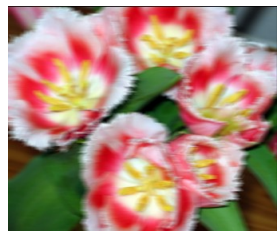
Blurred image



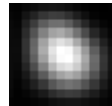
PSF estimation

Deconvolution

Image restoration flowchart



Blurred image



PSF estimation

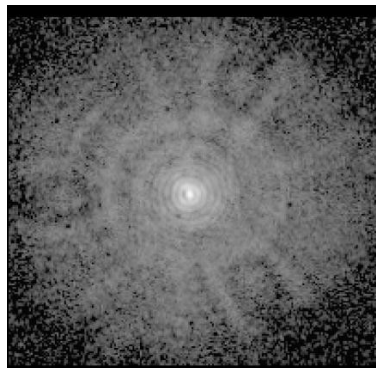
Deconvolution



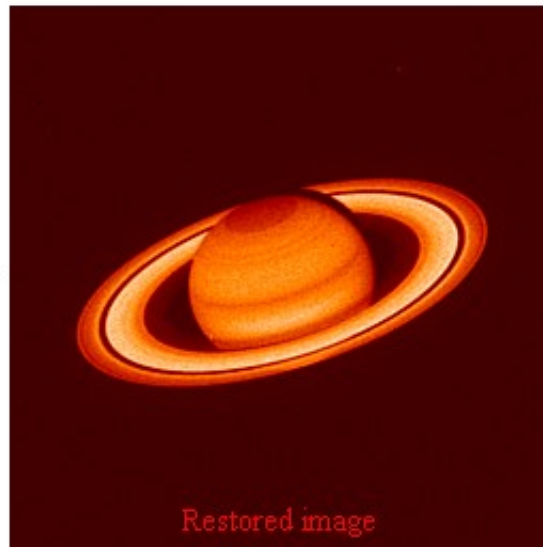
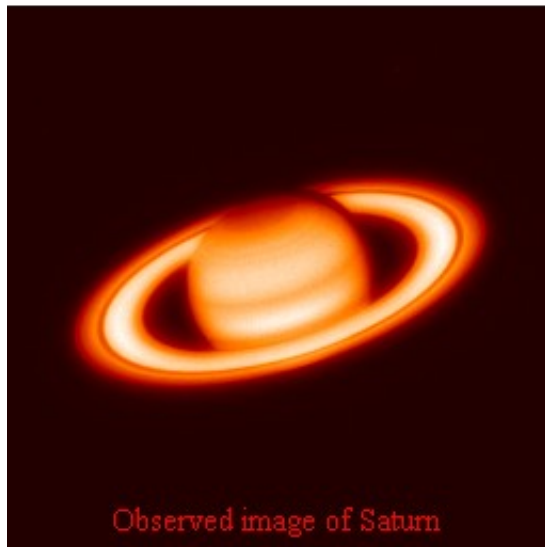
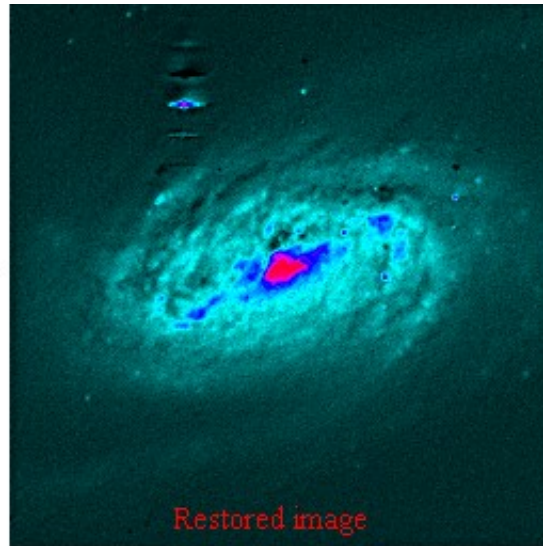
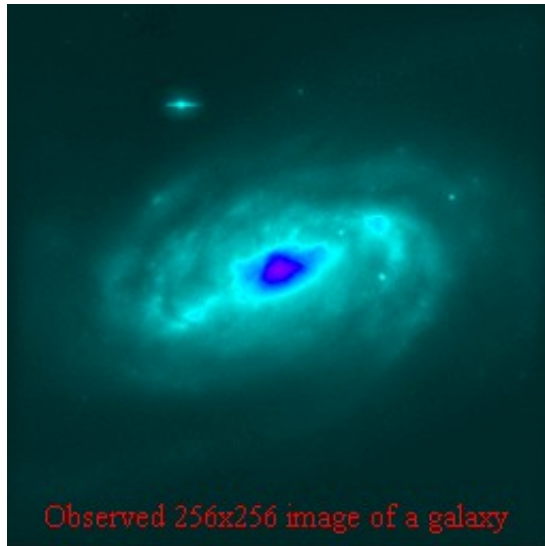
Restored image

$$h * u = \int u(x - s, y - t)h(s, t)dsdt$$

Images from the Hubble Space Telescope



Images from the Hubble Space Telescope



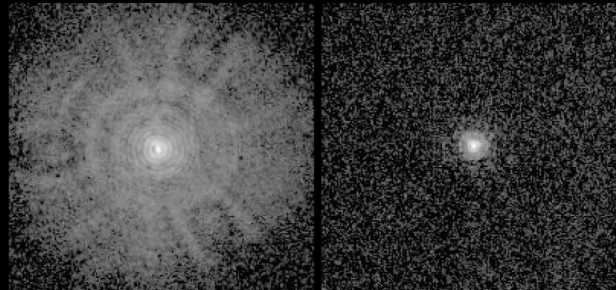
Hubble Space Telescope after COSTAR



Wide Field Planetary Camera 1



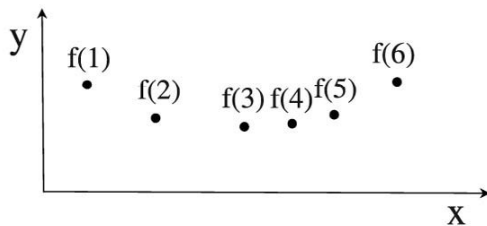
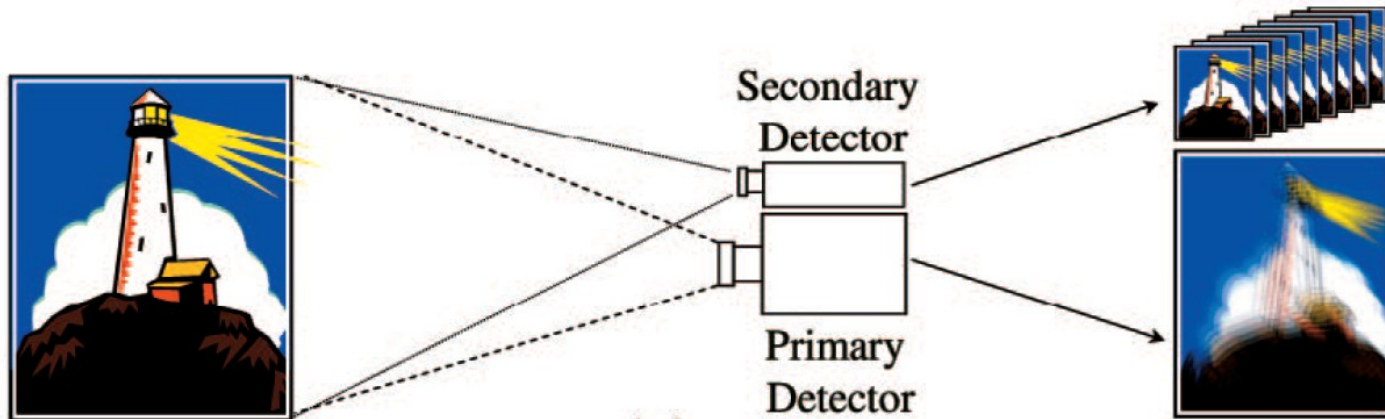
Wide Field Planetary Camera 2



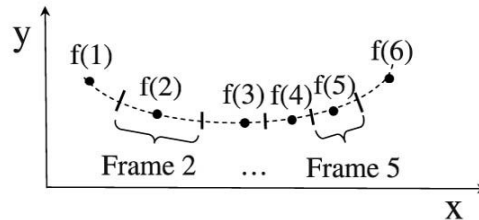
BEFORE COSTAR

AFTER COSTAR

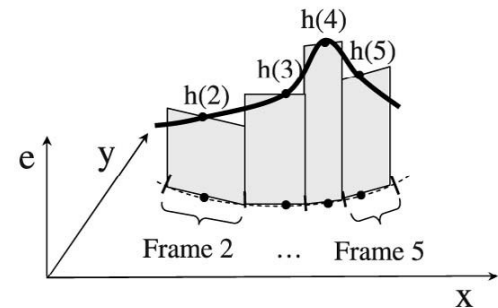
High-speed camera



**Sampled
trajectory
(secondary**



**Interpolated
trajectory**



**Estimated
PSF**

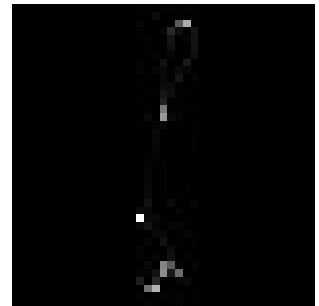
PSF estimation in smartphones



- Camera shake blur
- Dominant motion - rotation

PSF estimation in smartphones

- Using accelerometers and/or gyroscopes
- Rotation and translation of the phone



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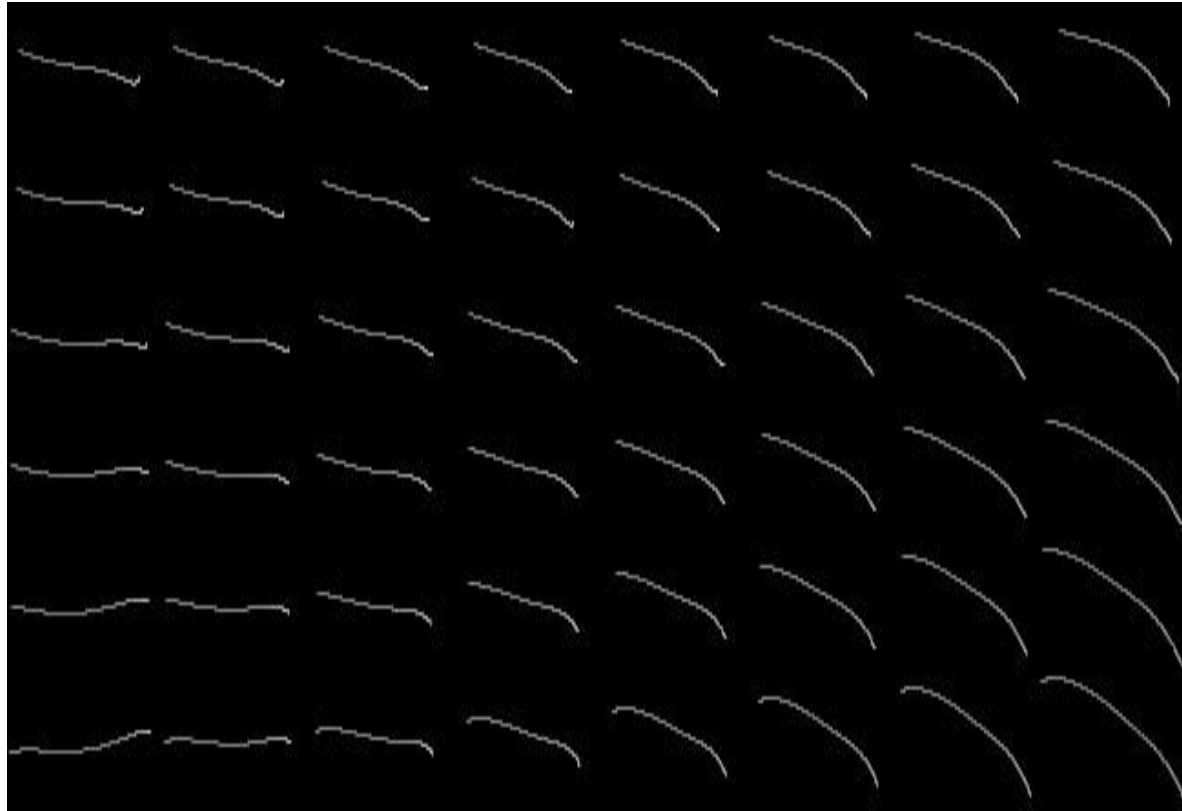


Acquired blurred image





PSF estimation

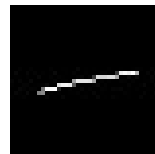
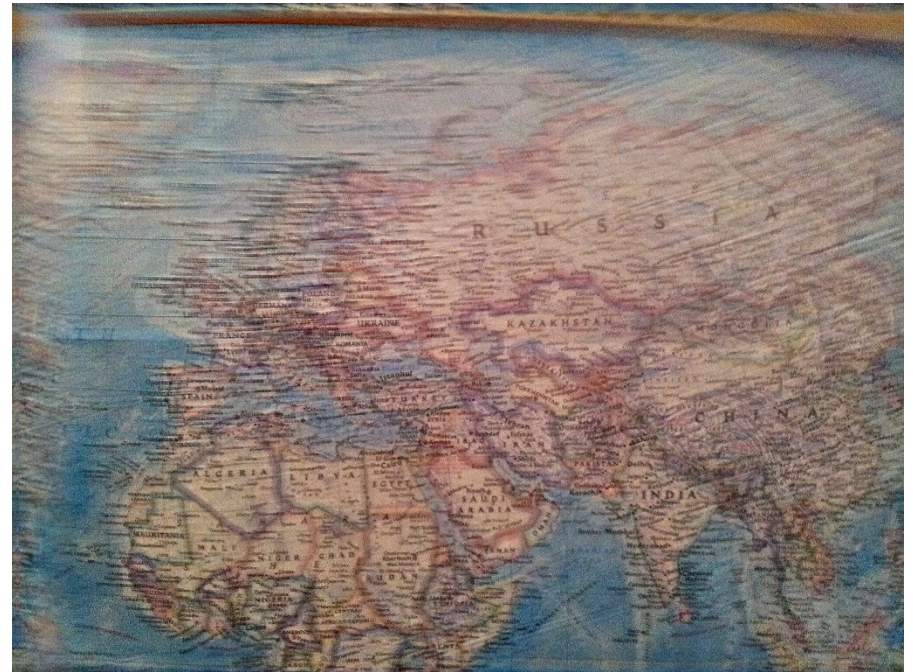




Wiener filtering



Wiener filtering in Samsung smartphone

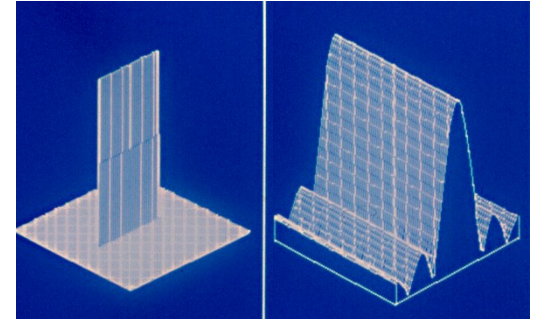


Restoration categories

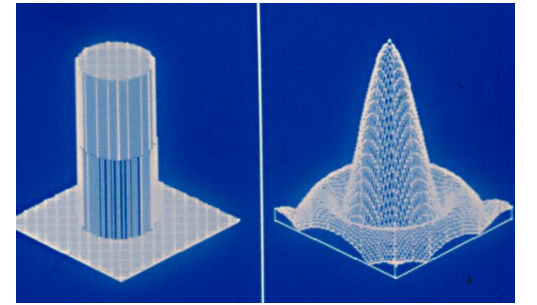
- PSF is completely known
- PSF is constant and of a known parametric shape
- PSF is constant and unknown
- PSF is variable and unknown

Common point-spread functions

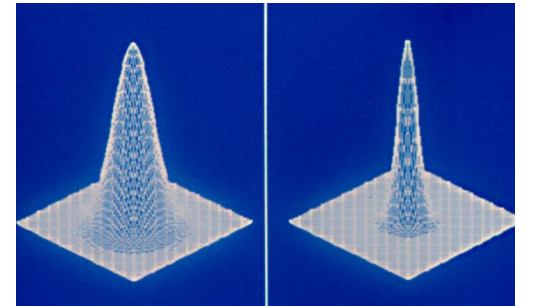
Motion blur: 1-D rectangular pulse, $FT = \text{sinc}(u)$

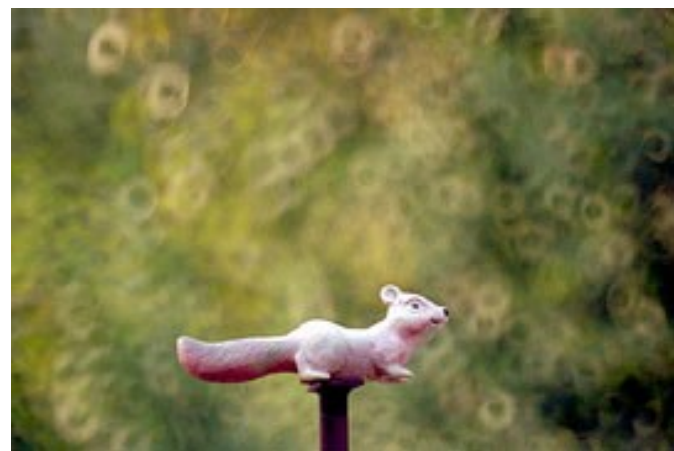
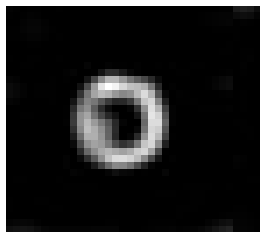
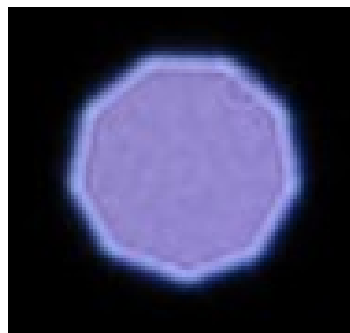
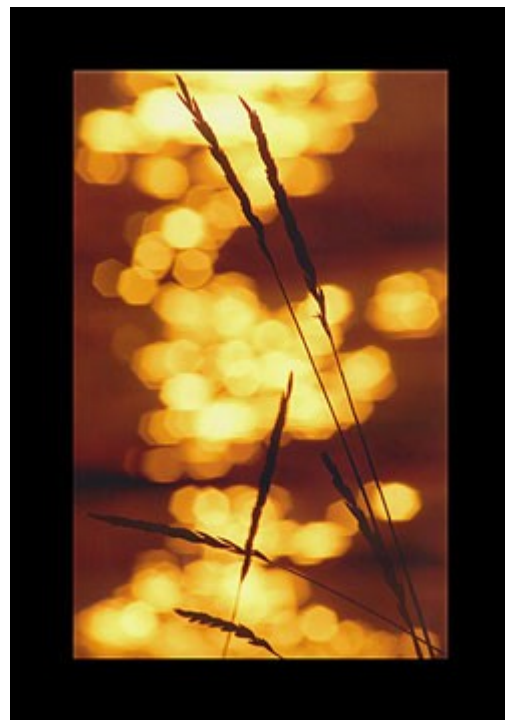
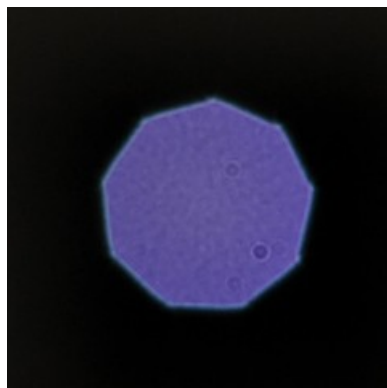
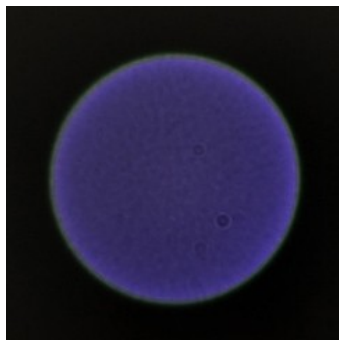


Out-of-focus blur: Cylinder, $FT = B(r)/r$



**Atmospheric turbulence: Gaussian $G(d)$,
 $FT = \text{Gaussian } G(1/d)$**

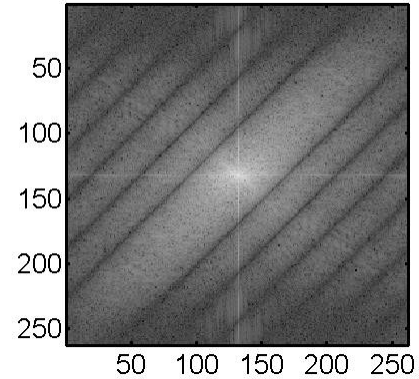
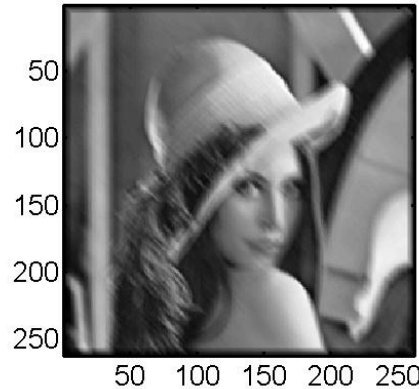




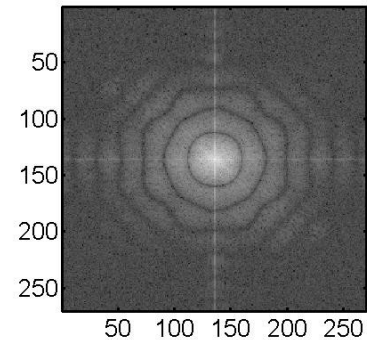
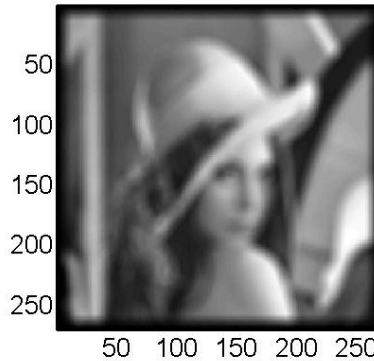


Spectrum of a degraded image

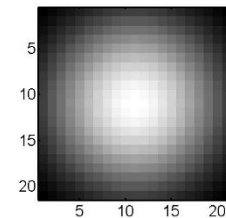
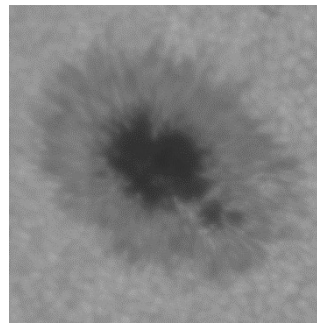
Motion blur



Out-of-focus blur



Atmospheric turbulence blur



Motion blur restoration by Wiener filter

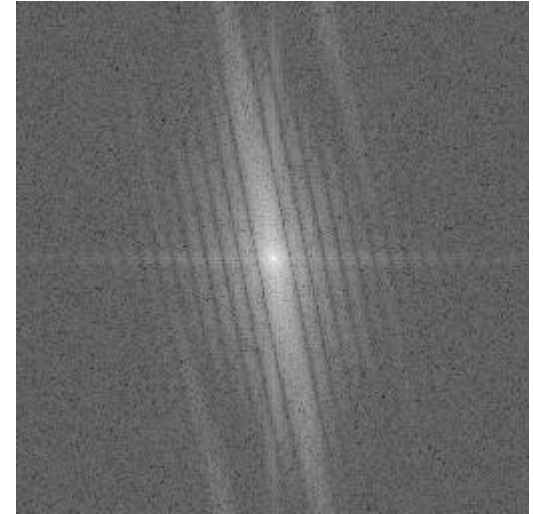
Original



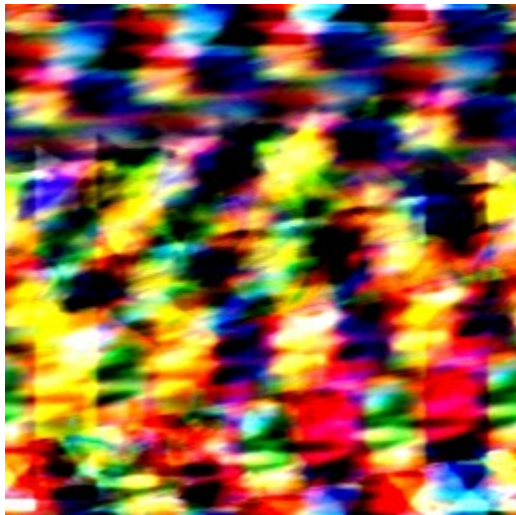
Motion blurred



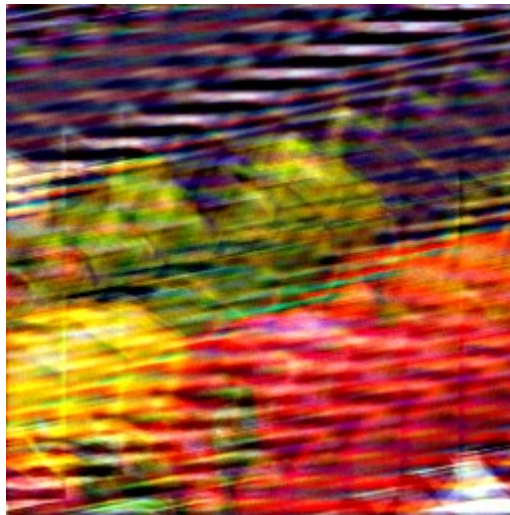
Spectrum



Wrong velocity



Wrong direction

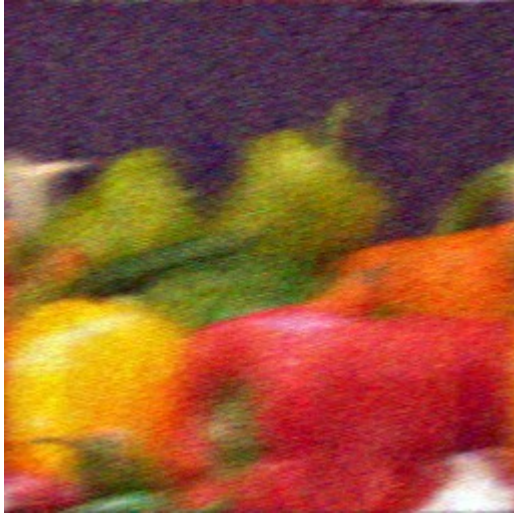


Correct PSF

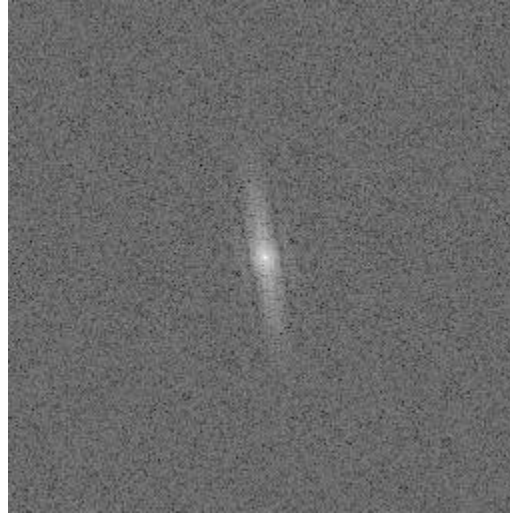


Motion blur restoration by Wiener filter

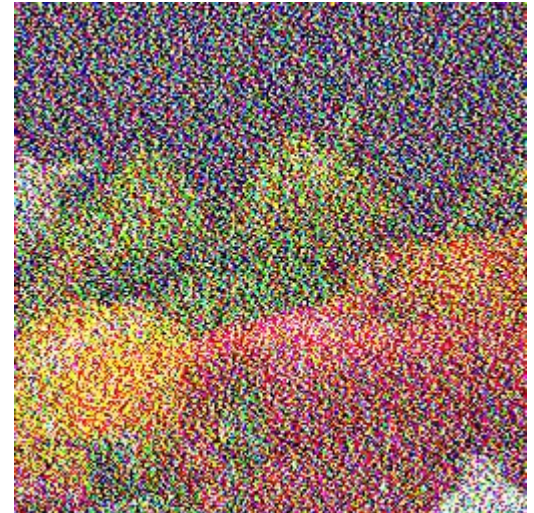
Blur + noise



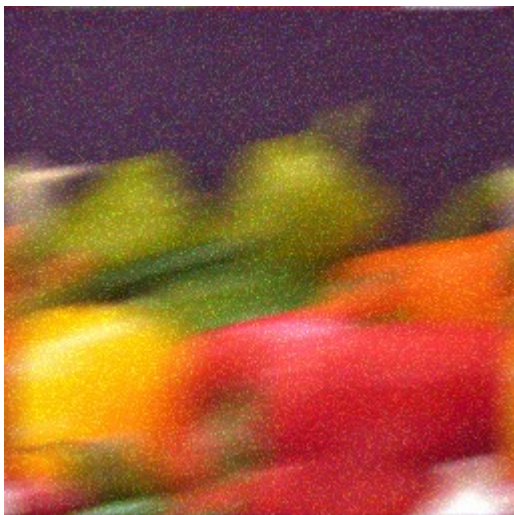
Spectrum



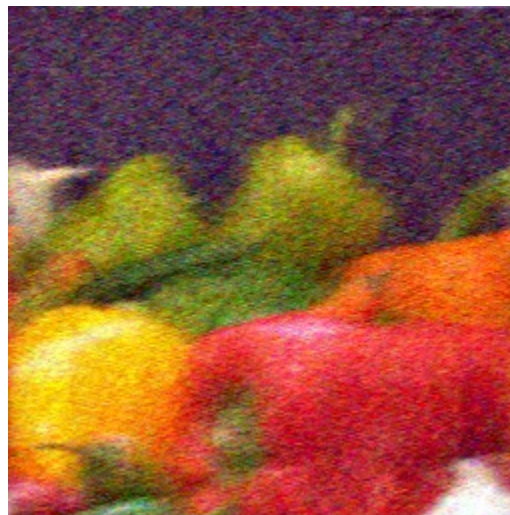
Inverse filtering



Constant SNR (good)

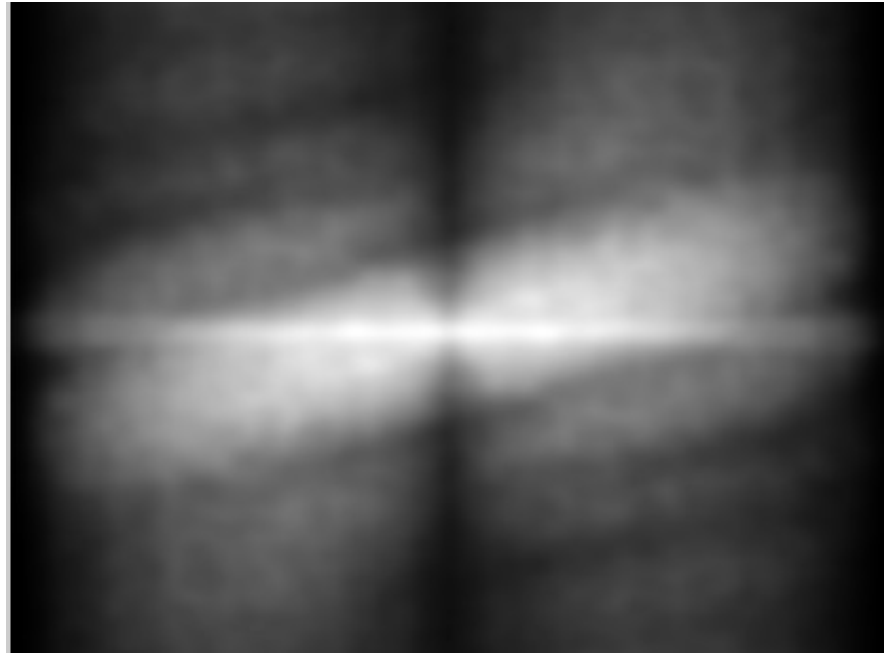


Constant SNR (wrong)



Actual SNR





Licence plate recognition



Restoration categories

- PSF is completely known
- PSF is constant and of a known parametric shape
- PSF is constant and unknown
- PSF is variable and unknown

Blind deconvolution

$$z(x) = (h * u)(x) + n(x)$$

- almost impossible to resolve
- solution ambiguity

$$z(x) = ((h_1 * h_2 * \cdots * h_L) * u)(x) + n(x)$$

Alternating Minimization

$$\min_{u,h} E(u, h) = \min_{u,h} \frac{1}{2} \|h * u - z\|^2 + \lambda Q(u) + \gamma R(h)$$

- Alternating Minimization

1. *u-step*: $\tilde{u} = \arg \min_u E(u, \tilde{h})$

2. *h-step*: $\tilde{h} = \arg \min_h E(\tilde{u}, h)$

3. *repeat 1 and 2.*

Example of blind deconvolution



Blurred image
 $z(x)$



Reconstructed image
 $\tilde{u}(x)$

Image blind deconvolution



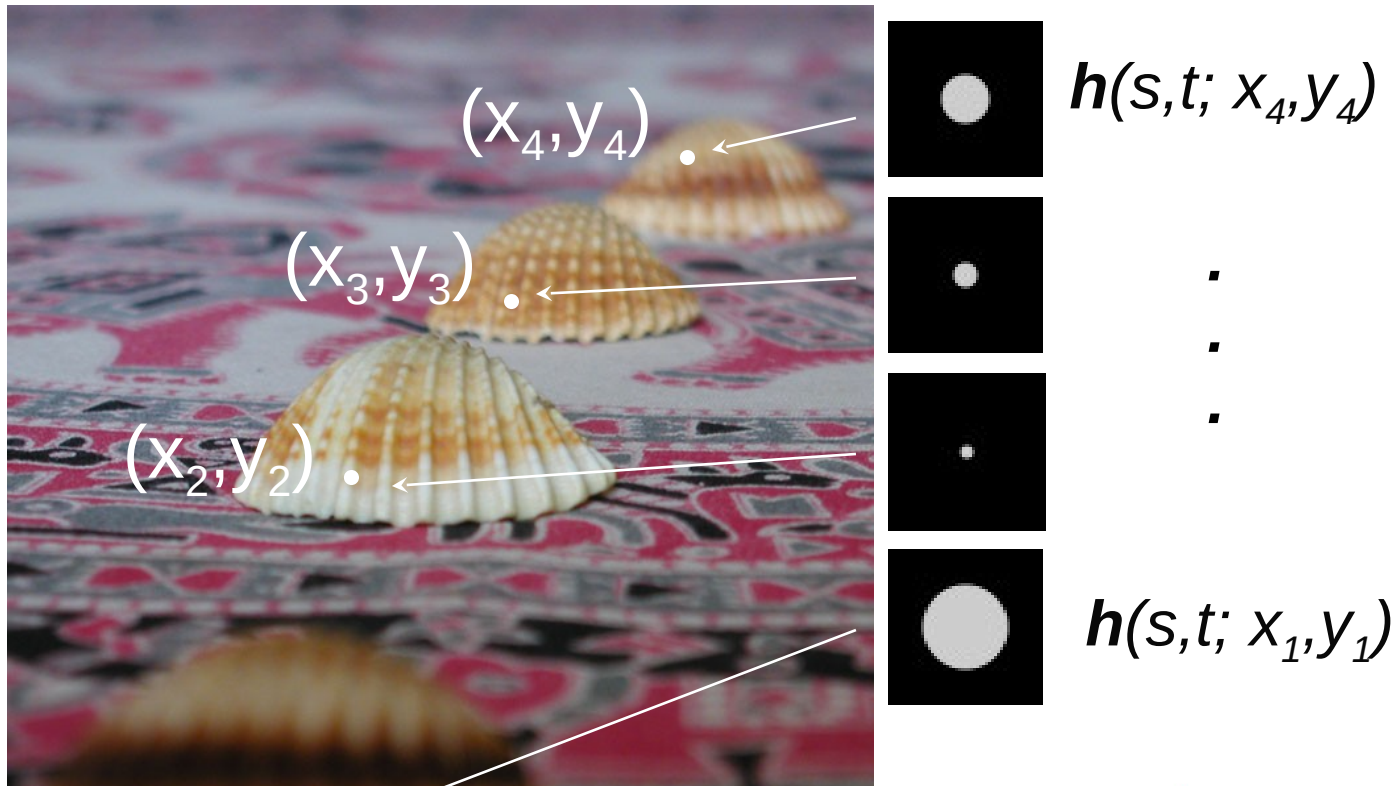
Restoration categories

- PSF is completely known
- PSF is constant and of a known parametric shape
- PSF is constant and unknown
- PSF is variable and unknown

Space-variant PSF



Space-variant PSF

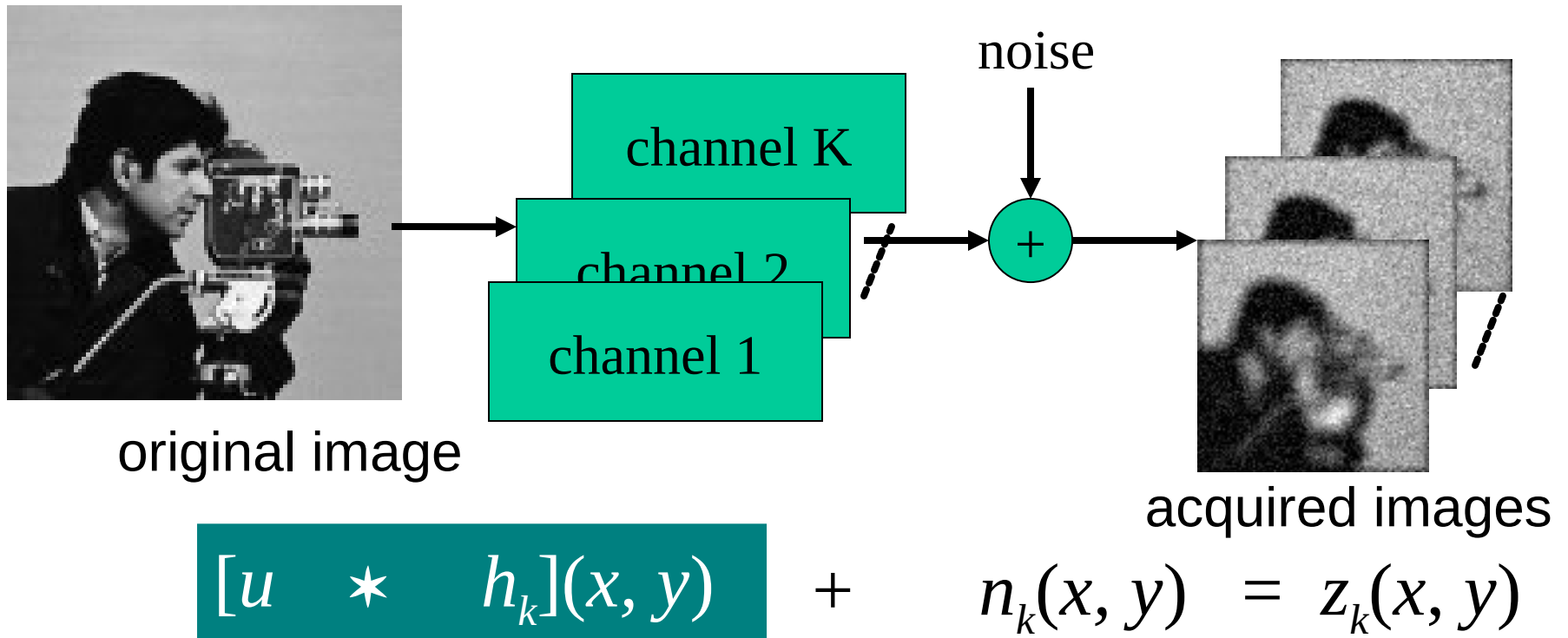


$$\mathbf{z}(x, y) = \int_{\Omega} \mathbf{u}(x - s, y - t) \mathbf{h}(s, t; x - s, y - t) ds dt + \mathbf{n}(x, y)$$

Multichannel image restoration

- Assumptions:
- Several input images of the same scene are available
- They are blurred by convolution with different convolution kernels
- The original scene does not change during the acquisitions

Multichannel acquisition model



MC Blind Deconvolution

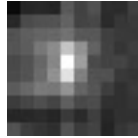
- System of integral equations
(ill-posed, underdetermined)

$$z_k(x) = (h_k * u)(x) + n_k(x)$$

- Energy minimization problem (well-posed)

$$E(u, \{h_i\}) = \frac{1}{2} \sum_{i=1}^K \|h_i * u - z_i\|^2 + \lambda Q(u) + \gamma R(\{h_i\})$$

Out-of-focus Camera



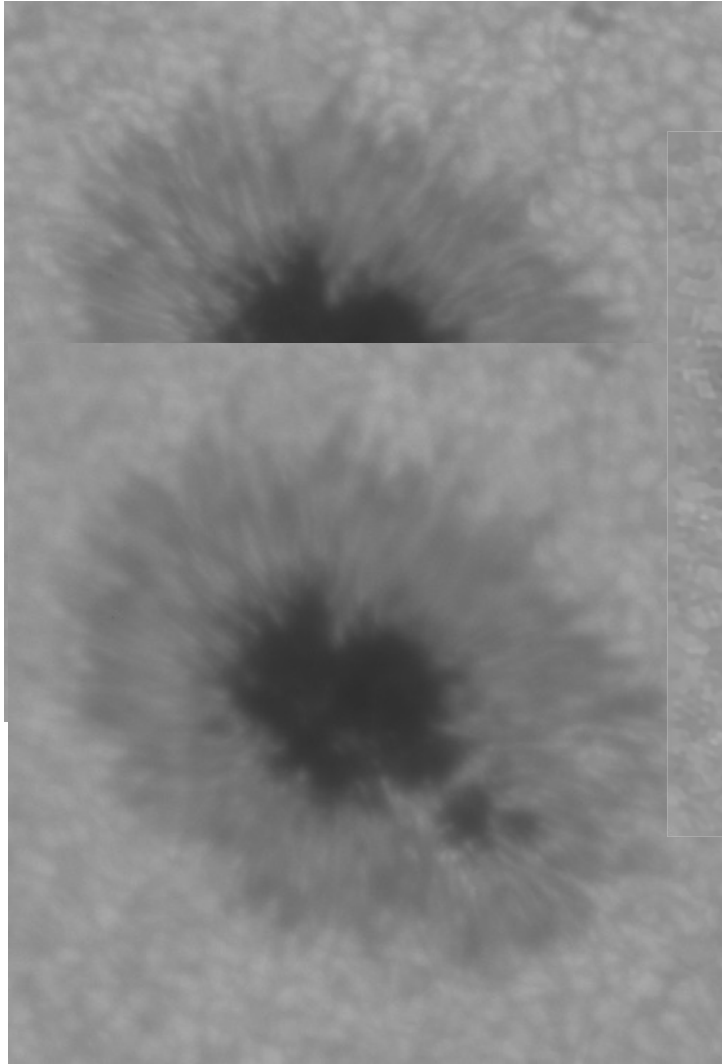
reconstructed



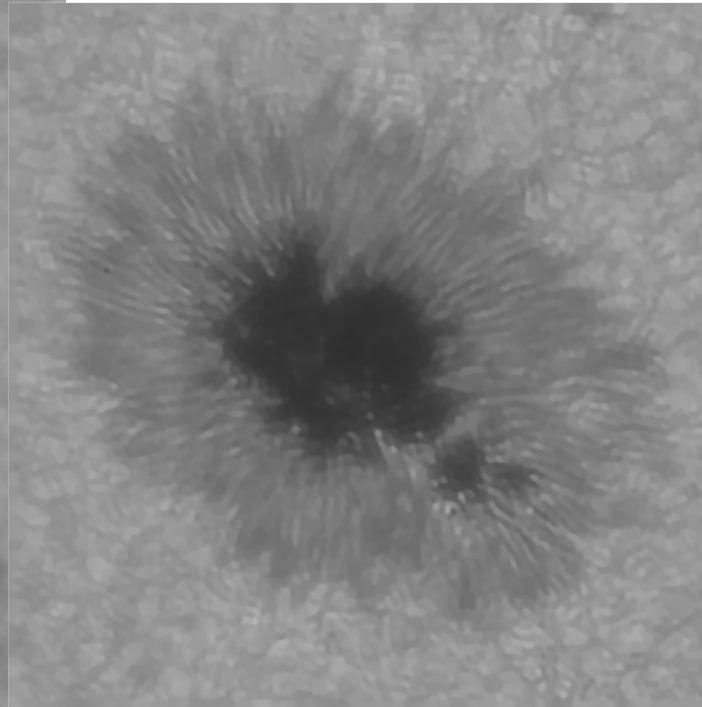
in focus

Astronomical images

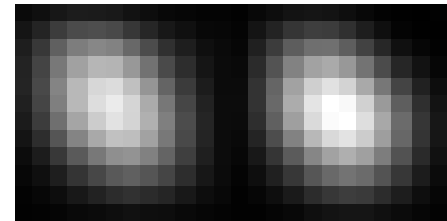
Degraded images



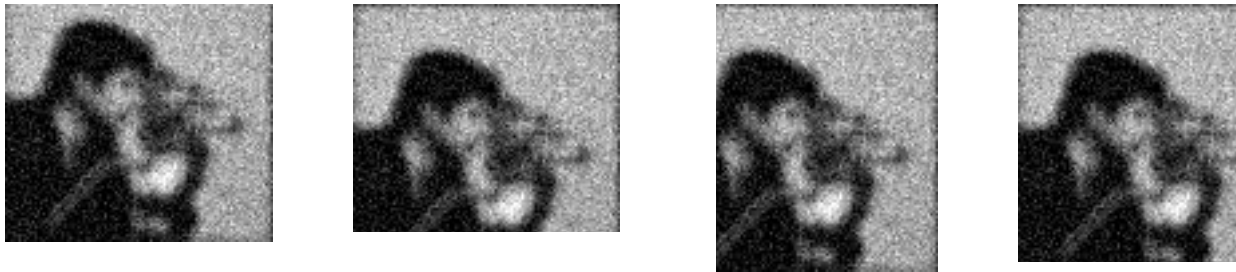
Reconstructed image



PSF estimation



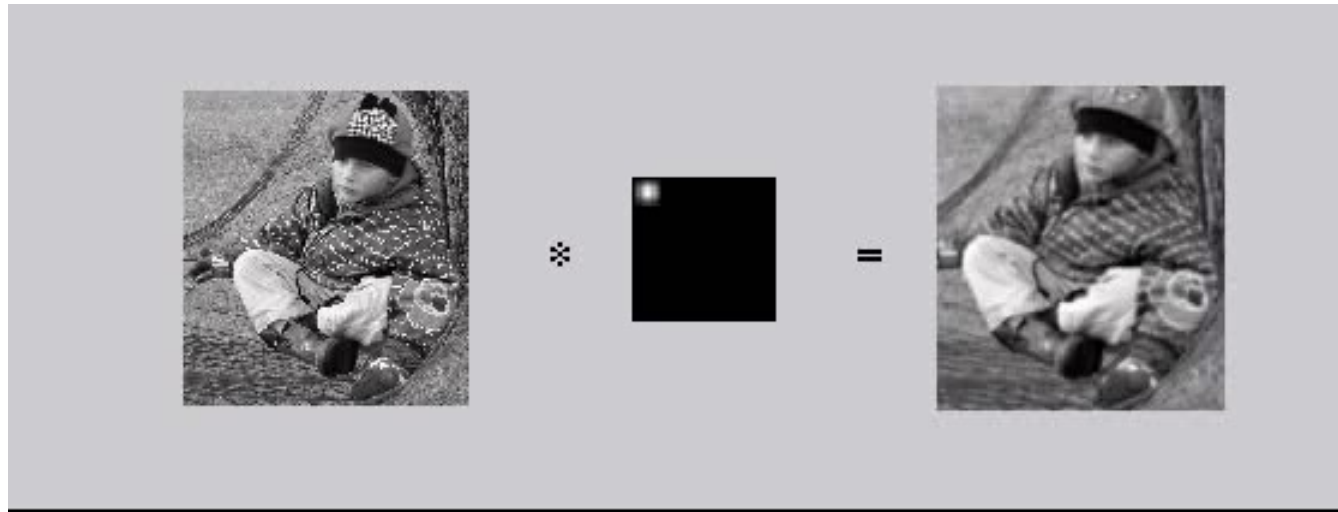
Misregistration of the channels



... leads to artefacts if not handled properly



Incorporating a between-image shift



original image

PSF

degraded image

$$[u \star h_k](\tau_k(x, y)) + n_k(x, y) = z_k(x, y)$$

$$[u \star g_k](x, y) + n_k(x, y) = z_k(x, y)$$

Simulated blurring



misalignment



misalignment
compensation

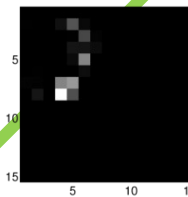
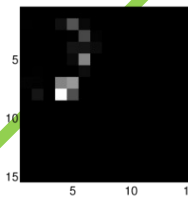
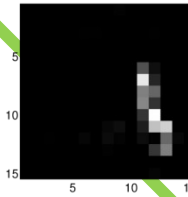


reconstructed image





Long-time exposure I





(a) Blurred input images, 1024×768 pixels



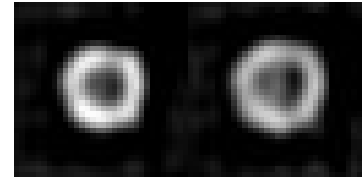
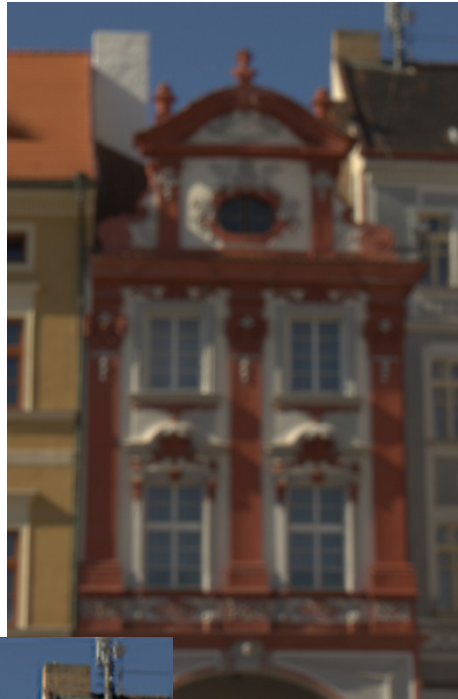
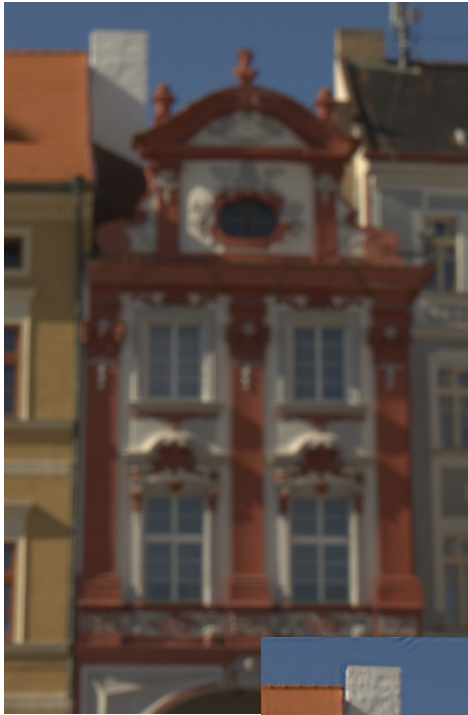
(b) Deconvolution from only first image



(c) Result of multi-image deconvolution

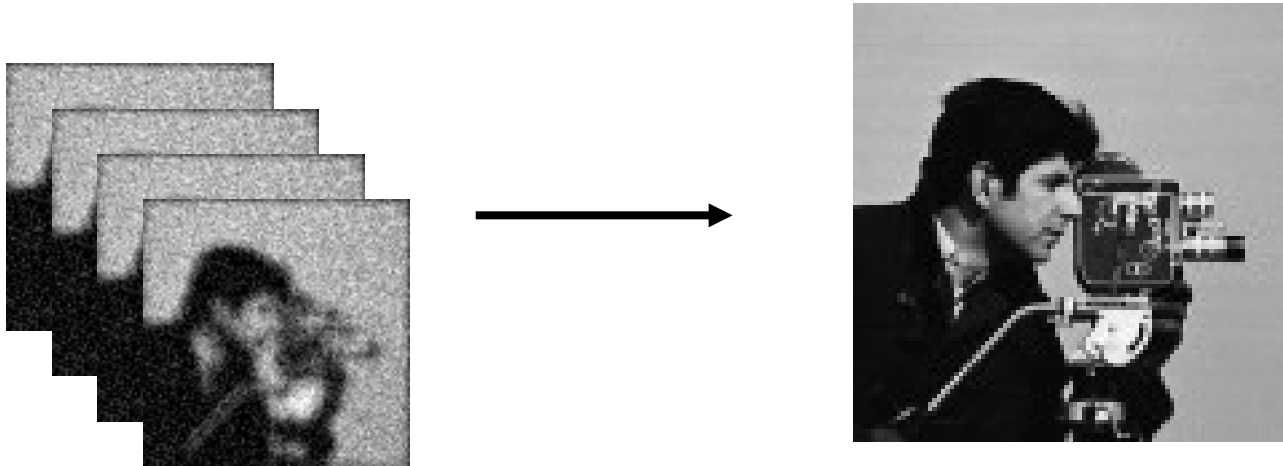
Out of focus blur





Super-resolution imaging

Method: Multiple acquisitions of the same scene with subpixel shift



Aliasing

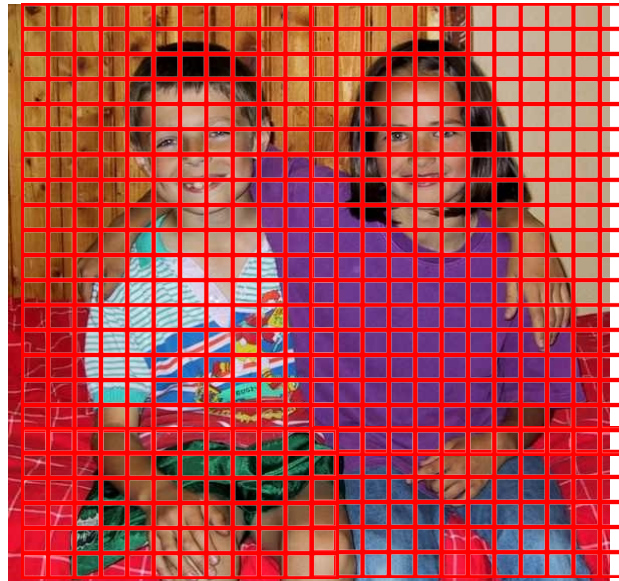
- The loss of the high frequencies (details) due to an insufficient resolution of the camera



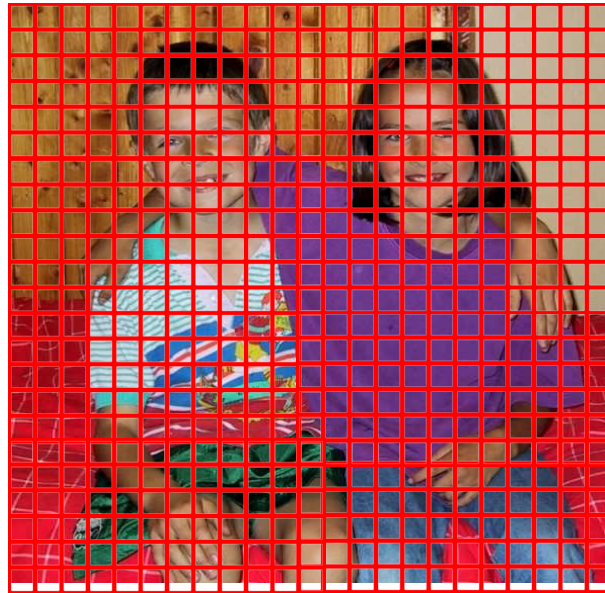
To suppress aliasing, apply multiple acquisitions with sub-pixel shifts



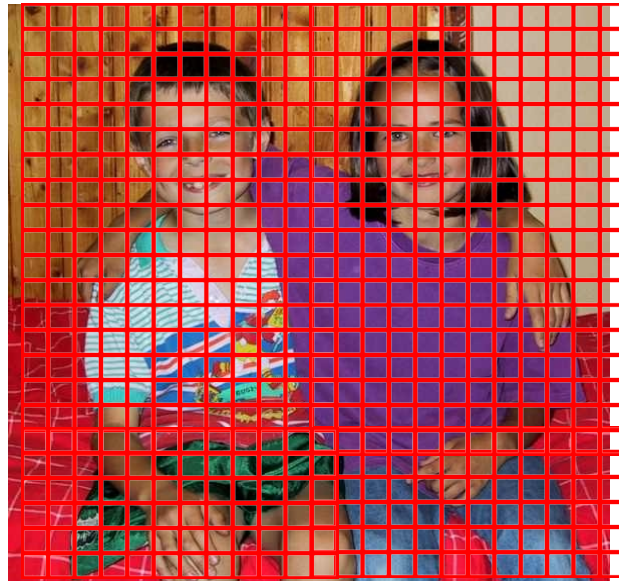
Multiple acquisitions with shifts



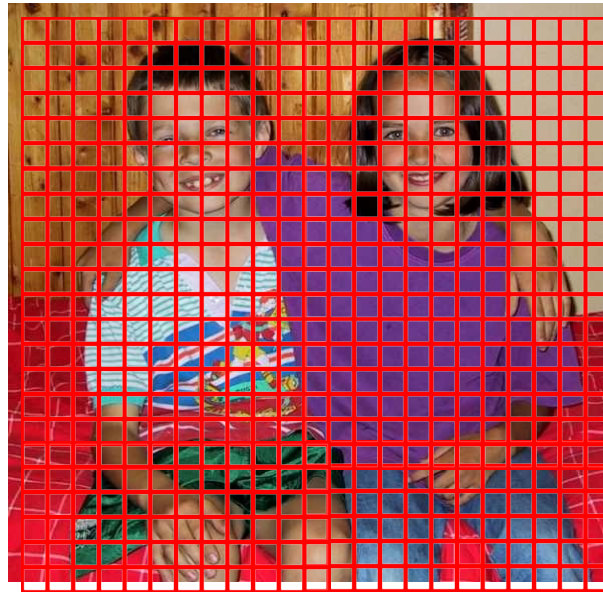
Multiple acquisitions with shifts



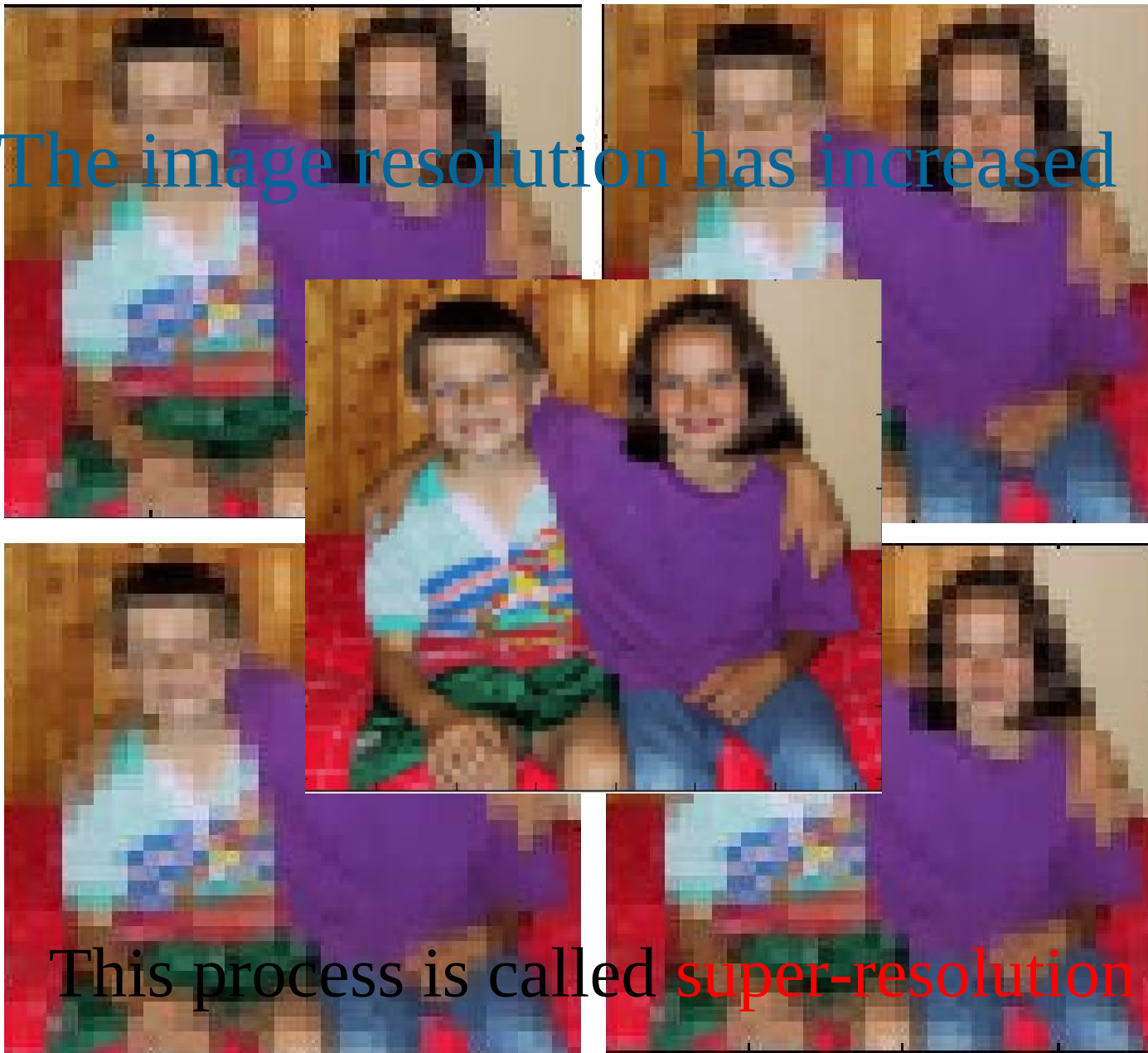
Multiple acquisitions with shifts



Multiple acquisitions with shifts



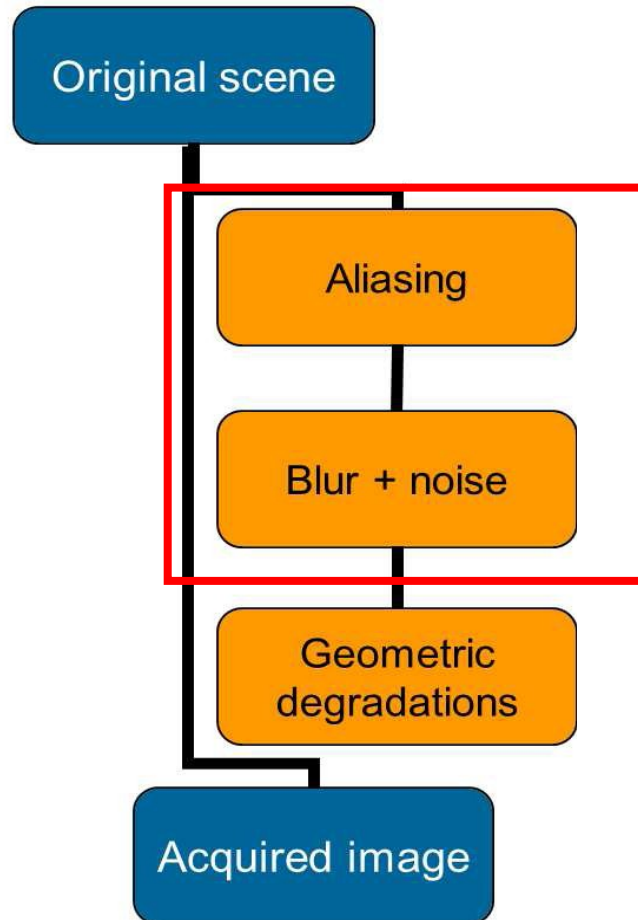
The image resolution has increased



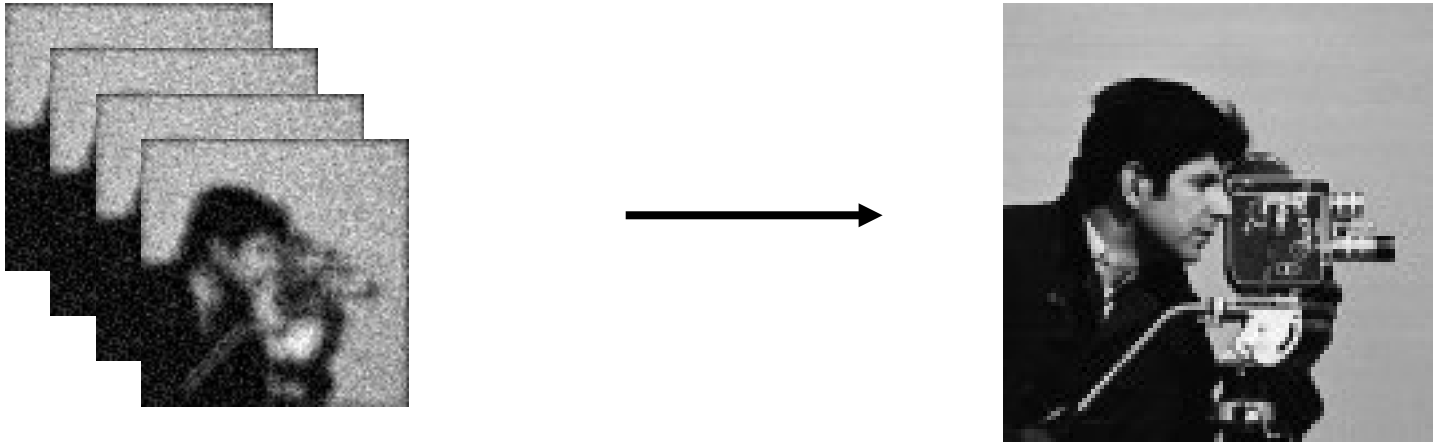
This process is called **super-resolution**

Realistic superresolution

SR must include also
de-blurring



Realistic superresolution



$$E(u, \{g_i\}) = \frac{1}{2} \sum_{i=1}^K \|D(g_i * u) - z_i\|^2 + \lambda Q(u) + \gamma R(\{g_i\})$$